Brouwer Fixed-Point Theorem via No Retraction

An End-to-End Meta-Math Walkthrough

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Abstract

A complete authored walkthrough of brouwer fixed-point theorem via no retraction, with explicit mathematical contracts, local verification gates, and reusable exports. The report presents the target theorem, declared imports, a dependency graph, five-component node contracts, a continuous proof, a worked example, and a hypothesis-repair audit. It is an authored mathematical exposition, not a deployed-agent transcript or a formal proof-assistant certificate.

AT-04 / Algebraic Topology / Report v1

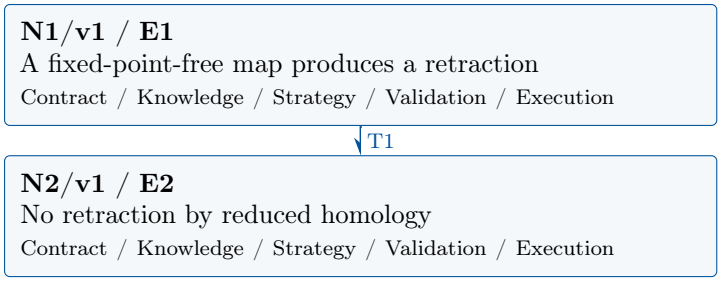
I. TARGET AND IMPORT CONTRACT

Theorem I.1. *For every $n \geq 1$ and continuous $f : D^n \to D^n$, prove f has a fixed point by deriving a forbidden retraction $D^n \to S^{n-1}$ from a fixed-point-free map.*

A. Foundations and permitted imports

D^n is the closed Euclidean unit ball and $S^{(n-1)}$ its boundary, $n \geq 1$. Import homotopy invariance of reduced singular homology, functoriality, contractibility of D^n, and the sphere computation $\widetilde{H}_{n-1}(S^{(n-1)};Z) = Z$. For $n = 1$ this is reduced $H_0(S^0) = Z$, while reduced $H_0(D^1) = 0$. The no-retraction result is proved below.

II. NODE INTERACTION WORKFLOW



Each box is one mathematical agent containing all five functional components. Blue arrows carry the versioned exports listed below. The layout gives a reading order; only drawn arrows assert dependencies. Diagnostic repair is separate from the proof DAG.

A. Versioned evidence-transfer ledger

T1: N1/E1/v1 to N2. A continuous *retractionr* : $D^n \to S^{(n-1)}$ under the no-fixed-point assumption.

Receiving activity. N2: No retraction by reduced homology

Compatibility. Use the same mathematical objects and hypotheses as the source export. Receiving contract: Retractionr and boundary inclusioni; prior exports E1/v1

III. AGENT ROLES AND FIVE-COMPONENT ACTIVITIES

A. N1/v1: A fixed-point-free map produces a retraction

Object and role. Extend the ray from $f(x)$ throughx to its last sphere intersection.

Inputs. *Continuous*$f : D^n \to D^n$ with $f(x) \neq x$ assumed for contradiction; foundational imports stated in §1

C — Contract

Own E1/v1: A continuous *retraction*$r : D^n \to S^{(n-1)}$ under the no-fixed-point assumption.

K — Knowledge

Retain *Continuous*$f : D^n \to D^n$ with $f(x) \neq x$ assumed for contradiction; foundational imports stated in §1; keep their assumptions and versions attached.

S — Strategy

Extend the ray from $f(x)$ throughx to its last sphere intersection.

V — Validation

Continuity and the boundary identity must both be proved.

E — Execution

Carry out the numbered deductions and publish E1/v1 together with its hypotheses and proof location.

Numbered mathematical activities

N1.1 Put $v = x - f(x) \neq 0$.

N1.2 The positive endpoint of the ray $y = f(x) + tv$ on the unit sphere is obtained at $t(x) = [-f(x) \cdot v + \sqrt{((f(x) \cdot v)^2 + (1 - \|f(x)\|^2)\|v\|^2)}]/\|v\|^2$.

N1.3 The denominator is nonzero.

N1.4 This is the larger root of $\|f(x) + tv\|^2 = 1$, and $t \geq 1$ because x at $t = 1$ lies in the ball.

N1.5 If $f(x)$ lies on the sphere then $f(x) \cdot v < 0$ unless $x = f(x)$, so the chosen root is still positive.

N1.6 The displayed expression is continuous, and $r(x) = f(x) + t(x)v$ is sphere-valued.

N1.7 Ifx is on the sphere, $t = 1$: the ball's ray intersection ends there, since distinct points of a Euclidean sphere have a strictly interior connecting segment.

N1.8 Thus $r(x) = x$ on the boundary, giving a retraction.

Export E1/v1. A continuous *retraction*$r : D^n \to S^{(n-1)}$ under the no-fixed-point assumption.

Acceptance gate. Continuity and the boundary identity must both be proved. Publish only after checking every displayed deduction.

Targeted repair. If the check fails, revise N1: Continuity and the boundary identity must both be proved. Recheck dependent nodes N2.

B. N2/v1: No retraction by reduced homology

Object and role. Apply functoriality in the one degree that separates disk and sphere.

Inputs. Retractionr and boundary inclusioni; prior exports E1/v1

C — Contract

Own E2/v1: Brouwer's fixed-point theorem in every stated dimension.

K — Knowledge

Retain Retractionr and boundary inclusioni; prior exports E1/v1; keep their assumptions and versions attached.

S — Strategy

Apply functoriality in the one degree that separates disk and sphere.

V — Validation

OrdinaryH $_0$ of a disk is Z; reduced homology is essential for the $n = 1$ version of this argument.

E — Execution

Carry out the numbered deductions and publish E2/v1 together with its hypotheses and proof location.

Numbered mathematical activities

N2.1 A retraction would satisfy $r \circ i = \mathrm{id}_S$.

N2.2 Hence $r_* \circ i_* = \mathrm{id}$ on $\widetilde{H}_{n-1}(S;Z) = Z$.

N2.3 But i_* lands in $\widetilde{H}_{n-1}(D;Z) = 0$, so the composite is the zero map.

N2.4 It cannot equal the identity on Z.

N2.5 This contradiction excludes the retraction, thus excludes a fixed-point-freef.

N2.6 Every continuous disk self-map has a fixed point, including $n = 1$.

Export E2/v1. Brouwer's fixed-point theorem in every stated dimension.

Acceptance gate. OrdinaryH $_0$ of a disk is Z; reduced homology is essential for the $n = 1$ version of this argument. Publish only after checking every displayed deduction.

Targeted repair. If the check fails, revise N2: OrdinaryH $_0$ of a disk is Z; reduced homology is essential for the $n = 1$ version of this argument. Withhold this terminal result until its gate passes.

IV. CONTINUOUS MATHEMATICAL PROOF

Read these steps in order as one proof; the diagram is optional. Numbered activities identify the specific constructions and deductions owned by each node.

A. §4.N1 — A fixed-point-free map produces a retraction

1. Put $v = x - f(x) \neq 0$.
2. The positive endpoint of the ray $y = f(x) + tv$ on the unit sphere is obtained at $t(x) = [-f(x) \cdot v + \sqrt{((f(x) \cdot v)^2 + (1 - \|f(x)\|^2)\|v\|^2)}]/\|v\|^2$.
3. The denominator is nonzero.
4. This is the larger root of $\|f(x) + tv\|^2 = 1$, and $t \geq 1$ because x at $t = 1$ lies in the ball.
5. If $f(x)$ lies on the sphere then $f(x) \cdot v < 0$ unless $x = f(x)$, so the chosen root is still positive.
6. The displayed expression is continuous, and $r(x) = f(x) + t(x)v$ is sphere-valued.
7. Ifx is on the sphere, $t = 1$: the ball's ray intersection ends there, since distinct points of a Euclidean sphere have a strictly interior connecting segment.
8. Thus $r(x) = x$ on the boundary, giving a retraction.

B. §4.N2 — No retraction by reduced homology

1. A retraction would satisfy $r \circ i = \mathrm{id}_S$.
2. Hence $r_* \circ i_* = \mathrm{id}$ on $\widetilde{H}_{n-1}(S;Z) = Z$.
3. But i_* lands in $\widetilde{H}_{n-1}(D;Z) = 0$, so the composite is the zero map.
4. It cannot equal the identity on Z.
5. This contradiction excludes the retraction, thus excludes a fixed-point-freef.
6. Every continuous disk self-map has a fixed point, including $n = 1$.

□

V. WORKED ORCHESTRATION

The constant map $f(x) = a$ with $\|a\| \leq 1$ fixes $x = a$. The affine *contraction*$f(x) = x/2$ also fixes 0. In either example the hypothetical retraction formula fails at its fixed point because $v = 0$, exactly the point the contradiction proves must exist.

A. Boundary and hypothesis audit

A rotation of S^1 by a nonzero angle has no fixed point, so a sphere self-map does not satisfy the disk theorem. The corresponding rotation of D^2 does fix 0. Validation retains both the disk domain and self-map condition rather than extending the claim to arbitrary compact spaces.

VI. ILLUSTRATIVE FAILURE AND REPAIR

The following is an authored diagnostic scenario. Changed assumptions define a new contract; they do not retroactively prove the original statement.

A. Failure and correction

This is an authored teaching trace, not a recorded agent run.

1. **Contract → Knowledge:** register the §1 theorem, imports, and requested exports before beginning the proof.
2. **Strategy → Execution:** choose the bridges in §3 and carry out the numbered activities in §4; transmit only established E-exports along the §2 DAG.
3. **Validation → responsible node:** test the specific boundary/failure in §5 and the node's individual audit. A failing candidate is withheld; downstream nodes may not use it.
4. **Knowledge + Strategy:** identify whether the error is an omitted condition, an invalid inference, or an attempted extension beyond the theorem. Restore the exact contract or explicitly state a different one.
5. **Execution:** redo the affected mathematical activity using the repaired condition or inference. Recheck consuming nodes, rather than feeding their conclusions back as premises.
6. **Contract:** publish only the supported export with its assumptions and proof location. A hypothetical revised export is v2; the written proof here is the v1 exposition, not a claim that this repair was executed.

B. Hypothesis boundary

A rotation of S^1 by a nonzero angle has no fixed point, so a sphere self-map does not satisfy the disk theorem. The corresponding rotation of D^2 does fix 0. Validation retains both the disk domain and self-map condition rather than extending the claim to arbitrary compact spaces.

C. Version and recheck

Treat any changed assumption as a new contract. Recheck the revised export and all of its descendants. A backward diagnostic never establishes an earlier theorem. This is an authored teaching trace, not an execution log.

Provenance. This authored proof is complete relative to its declared imports. Reuse each export with its assumptions, proof, and version. The original Markdown remains available for comparison.

Further reading.

- Allen Hatcher, Algebraic Topology: covering spaces, cellular homology, and cap products.